AMR as a promising automatic evaluation metric for machine translation

Xiao Wang

xiao.wang@student.uni-tuebingen.de

Abstract

This paper explores the use of Abstract Meaning Representation (AMR) as a novel metric for evaluating machine translation (MT) quality. Unlike traditional string-based metrics and embedding-based neural metrics, AMR captures deeper semantic structures by representing sentence meaning as graphs of concepts and relations. To assess its effectiveness, AMR-based evaluation scores are computed using Smatch, which measures similarity between the AMR graphs of system outputs and reference translations. The experiments are conducted on the WMT 2023 Metrics Shared Task dataset for two language pairs translating into English: Hebrew→English (he→en) and Chinese→English (zh→en). Results show that, when using high-quality synthetic reference translations, AMR achieves a global average correlation of 0.722 with human ratings, outperforming all string-based metrics and approaching the performance of several strong embedding-based metrics, such as *BERTScore* and *XLsim*. These findings indicate that AMR provides complementary semantic information overlooked by surface-level metrics. Moreover, since the reported results are obtained using minimal AMR parsing settings, further performance gains are expected with improved parsing techniques and document-level semantic modeling.

1 Introduction

Automatic evaluation metrics (AEM) for machine translation (MT) have evolved alongside advancements in MT systems, whether statistical or neural. These metrics typically involve three components as following:

(1) The **source text** of the translation generated by the machine translation system.

(2) The **translation hypothesis** of the source text generated by the machine translation system for evaluation.

(3) One **reference translation** of the same source text produced by humans.

The purpose of evaluation metrics is to assign a score that reflects the similarity between a machine-generated translation (the hypothesis) and a human-produced reference translation. Recently, emerging neural metrics have introduced a new approach by directly comparing machine translations with their corresponding source texts (tracing back to WMT19[1]). This method reduces the reliance on human-generated reference translations, saving both time and cost and avoiding reference-bias. Ultimately, the primary goal of any evaluation metric is to align its scores as closely as possible with human judgments. Because the better a metric correlates with human evaluation of the machine-generated translations, the more accurately it can assess their translation quality.

Automatic evaluation metrics are typically divided into two categories: traditional statistical metrics and state-of-the-art (SOTA) neural metrics. Traditional metrics assess the similarity between two strings—the machine-generated translation (hypothesis) and the human-produced reference—based purely on their textual content. These metrics often rely on comparing n-grams (sequences of words or characters) between the hypothesis and the reference, as seen in popular metrics like BLEU (Bilingual Evaluation Understudy) and chrF (character n-gram F-score). Others, such as WER (Word Error Rate), measure the num-

[1]WMT is the abbreviation for "Workshop on Machine Translation" , for details, see https:// machine-translate.org/wmt

ber of edits needed to convert the machine output into the reference, while metrics like TER (Translation Edit Rate) estimate the amount of effort a human-translator would need to post-edit the machine translation. Traditional metrics remain widely used—and often preferred—thanks to several clear advantages. They rely on efficient string-matching algorithms, making them computationally inexpensive. Their simplicity also makes the resulting scores easy to explain and reproduce, and they are generally language-independent. However, a major limitation of these metrics is their weak correlation with human judgments (check WMT Metrics Shared Task since 2021), which undermines their reliability in accurately assessing translation quality.

Neural metrics, on the other hand, demonstrate a strong alignment with human judgments (see WMT evaluations), particularly the fine-tuned ones, largely due to their use of powerful state-of-the-art neural networks. Unlike traditional metrics that rely on exact string matches, neural metrics incorporate contextual information and are often trained on human assessment data. As a result, they can recognize semantically similar translations, allowing for a more nuanced and accurate assessment of translation quality—one that aligns closely with human evaluations. Additionally, they can be fine-tuned on specific training data, further improving their correlation with human evaluations in targeted use cases. Despite their clear advantages, neural metrics have yet to gain dominance in the machine translation community (Marie et al., 2021) (although they are gradually gaining attention since 2021), largely due to equally significant disadvantages. They are considerably slower to compute than traditional metrics and often require substantial memory and GPU resources, especially when large language models (LLMs) are used. Their complexity—often involving millions or even billions of parameters—also makes the resulting scores difficult to interpret. Additionally, maintaining consistency and reproducibility can be challenging, as older implementations may become incompatible over time due to changes in software frameworks or insufficiently specified hyperparameters in the original research.

This paper aims to contribute to the exploration of novel machine translation (MT) evaluation metrics. It is motivated by recent research demonstrating the benefits of incorporating semantic representations—such as verb-argument relations and Abstract Meaning Representation (AMR)—into neural machine translation models (Song et al., 2019). Instead of using AMR as additional semantic features within MT models, this work investigates the potential of using AMR directly as a MT evaluation metric. Specifically, AMR-based evaluation measures translation quality by comparing the semantic representations of system outputs with those of human reference translations, or potentially even with the source text if multilingual AMR parsing with high accuracy becomes available. This can be a potential approach that bridges the gap between traditional string-matching metrics and computationally expensive neural metrics, offering a semantic-aware alternative with lower computational costs.

The goal of this study is to evaluate AMR as a novel MT evaluation metric and compare its performance against leading metrics reported in the WMT23 Metrics Shared Task to assess its viability. To ensure comparability and reliability, the same test data and system outputs from WMT23 are used. Due to resource constraints, the evaluation focuses on two language pairs translating into English: Hebrew to English (he→en) and Chinese to English (zh→en).

The paper begins with an introduction to AMR, followed by a description of the test data, systems, and benchmark metrics from the WMT23 Metrics Shared Task. It then presents the implementation of the AMR-based evaluation metric within the WMT evaluation framework and reports the corresponding results. Finally, the paper concludes with a discussion of findings and future directions.

2 AMR as a MT Evaluation Metric

In this section, I start with a general introduction to the Abstract Meaning Representation (AMR) framework, followed by a description of the AMR parser (text-to-AMR) used in this study. Finally, the limitations of AMR are dis-

cussed as potential drawbacks to its use as a semantic similarity metric to evaluate translation quality.

2.1 AMR Framework

As the name suggests, Abstract Meaning Representation (AMR) is a semantic representation framework that abstracts away from syntax to capture the underlying meaning of sentences. It represents this meaning using rooted, labeled, directed, and acyclic graphs where leaves represent concepts and edges show relations between two nodes—serving as a semantic counterpart to syntactic treebanks. As illustrated in the example in figure 1, in the sentence "*The boy wants to go.*", "want", "boy" and "go" are the concepts in AMR graph where "want" serves as the root, with "boy" and "go" as its arguments. These graph structures are designed to be both human-readable and machine-processable, making AMR a valuable resource for tasks in natural language understanding (NLU) and natural language generation (NLG). And the AMR framework has been developed primarily for English.

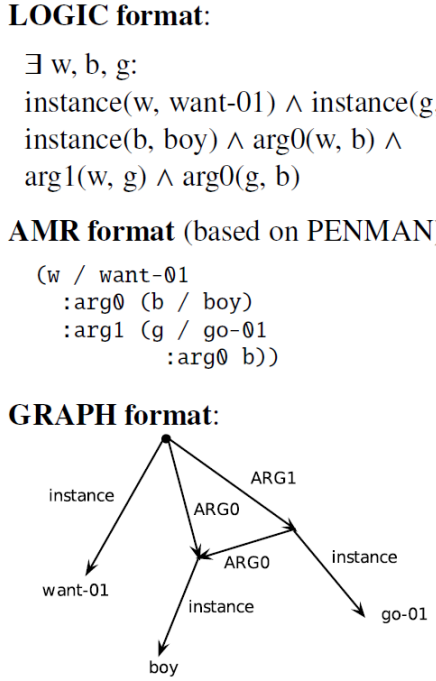


Figure 1: Equivalent formats for representating the meaning of "The boy wants to go" (Banarescu et al., 2013).

AMR was originally introduced by Langkilde and Knight in 1998 as part of the Nitrogen language generation system (Langkilde and Knight, 1998). Its use of labeled, directed graphs was inspired by the PENMAN Sentence Plan Language. However, AMR gained significant attention much later, in 2013, when Banarescu et al. substantially expanded the framework by publishing 50 pages of annotation guideline. Their work aligned AMR with PropBank[2] and enriched it with additional semantic features such as non-core semantic roles (e.g., *time, place, manner*), negation, modality, quantities, questions, inverse relations and relations among multiple predicates (Banarescu et al., 2013). Since its initial release in 2013, the AMR framework has undergone two major updates. The latest version in 2020, AMR Annotation Release 3, includes hundreds of newly added or refined predicate-argument frames, improved guideline quality control, and—most notably—an expanded AMR bank. The number of AMR-annotated sentences has increased from 39,260 to 59,255, with a subset of 8,027 sentences annotated at the document level to capture discourse phenomena such as coreference, implicit role references, and bridging relations. The sources of the sentences are detailed in Table 1.

Data Source	Training	Dev	Test	Total
BOLT DF MT (bolt)	1061	133	133	1327
Broadcast conversation (cctv)	214	0	0	214
Weblog and WSJ (consensus)	0	100	100	200
BOLT DF English (dfa)	7379	210	229	7818
DEFT DF English (dfb)	32915	0	0	32915
Aesop fables (fables)	49	0	0	49
Guidelines AMRs (guidelines)	970	0	0	970
LORELEI (lorelei)	4441	354	527	5322
2009 Open MT (mt09sdl)	204	0	0	204
Proxy Reports (proxy)	6603	826	823	8252
Weblog data (wb)	866	0	0	866
Wikipedia (wiki)	192	0	0	192
Xinhua MT (xinhua)	741	99	86	926
TOTAL	55635	1722	1898	59255

Table 1: Dataset statistics used in AMR Annotation Release 3

[2]PropBank is a human-annotated corpus of framesets. A frameset is a predicate (verbal proposition) and its arguments. For details please see https://propbank.github.io/

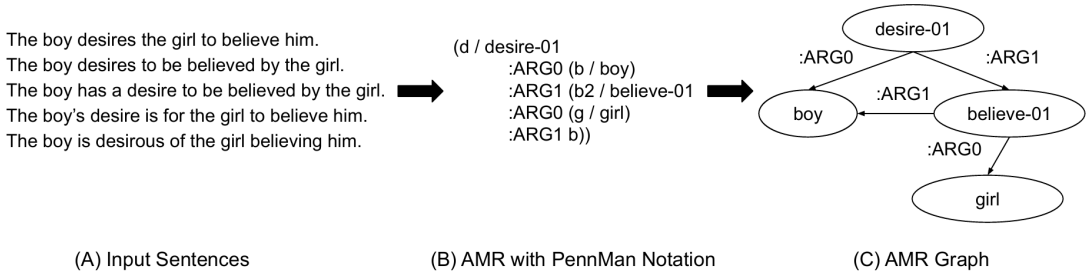


Figure 2: One Abstract Meaning Representation for five different input sentences (Mansouri, 2025)

Given this general understanding of AMR, several key features make it a promising candidate for evaluating the semantic similarity between two translations of the same sentence:

- It **captures the semantic structure of sentences** through the use of predicate-argument frames. Framesets such as *desire-01* define roles like the wanter (:ARG0) and the wanted (:ARG1), effectively revealing the underlying meaning of "who is doing what to whom" for each sentence. Additionally, it uses other non-core semantic relations to capture connections between entities, events, and properties—such as time, place, and manner. It also resolves coreference by reusing variables.
- It **normalizes over syntactic structures**. The use of semantic relations abstracts AMR away from surface syntax, meaning that sentences with different word orders or syntactic constructions —but conveying the same meaning—will yield the same AMR representation. For example, active and passive voice constructions of the same sentence will share the same AMR representation, regardless of their syntactic differences as shown in Figure 2.

These core features of AMR—such as its abstraction from surface syntax and its use of semantic relations—allow for the generation of semantic representations that focus on meaning rather than form. As a result, when applied to machine-generated translations, AMR can produce graph representations of machine-generated translations that closely align with those of their corresponding human reference translations if the machine translation correctly conveys the meaning of the source sentence, even if the surface wording differs. This makes AMR particularly well-suited for evaluating the semantic similarity between translations of the same source sentence. Furthermore, it also suggests that if multilingual AMR parsing becomes sufficiently robust and widely available, AMR could potentially be used to directly assess the semantic similarity between machine-generated translations and the original source sentences, thereby eliminating the need for reference translations. Moreover, with the growing availability of document-level AMR annotations, AMR could also be extended to capture cross-sentence semantic relations, enabling future evaluations of translation quality at the document level.

2.2 AMR Parser and Smatch Metric

Similar to syntactic parsing with treebanks, a natural progression in AMR development is the creation of automatic semantic parsers trained on annotated AMR corpora (semantic banks). These parsers are used to automatically annotate new data and help expand the AMR resource. In this study, the SOTA transition-based AMR parser from IBM[3], trained on the AMR Annotation Release 3 semantic bank (AMR3), is used to parse the English translations of the source sentences, converting each sentence into its AMR representation.

In order to evaluate parser accuracy as well as assess inter-annotator agreement (IAA) during annotation, two members of Banarescu et al.(Cai and Knight, 2013) also developed the ***Smatch*** metric—an evaluation tool that measures the similarity (or "semantic overlap" as Cai and Knight call it) between two AMR

[3]https://github.com/IBM/transition-amr-parser

graphs by comparing their sets (or conjunction) of logical triples. Examples of such triples are shown in Figure 1 in logical format. As we can see from Figure 1, the triples take one of two forms: either *relation(variable, concept)*, consisting of the elements "relation", "variable", and "concept"—as in *instance(w, want-01)*; or *relation(variable1, variable2)*, consisting of "relation", "variable1", and "variable2"—as in *arg0(w, b)*. The Smatch score between the AMR graphs of two sentences ranges from 0 to 1, with higher scores indicating greater semantic overlap or closer semantic equivalence. It is computed as an F-score, as follows:

$$\mathbf{F = 2*(P*R)/(P+R)}$$
with precision $\mathbf{P = M/T}$
with recall $\mathbf{R = M/G}$

where **M** is the number of matching triples, **T** is the total number of triples in the first AMR and **G** is the total number of triples in the second AMR.

In practice, since different variables may be used to represent the same concepts, there are often multiple possible ways to align (or map) the variables between two AMR graphs. Therefore, the Smatch algorithm performs a search over possible mappings and selects the one that yields the highest F-score, representing the maximum semantic overlap. This optimal F-score is then reported as the final Smatch score. For example, if we compute the Smatch score between the AMR of the sentence "The boy wants to go." (from Figure 1) and the AMR of the sentence "The boy wants a toy." represented as a conjunction of three logical triplets:

$$instance(x,\text{want-01}) \wedge instance(y, boy) \wedge instance(z, toy) \wedge arg0(x, y) \wedge arg1(x, z)$$

we obtain six possible mappings of variables between the two AMRs as shown in Table 2. Naturally, more efficient methods are employed to search for the optimal alignment, rather than exhaustively enumerating all possible variable mappings which becomes computationally expensive as the number of variables increases. For details on these optimization techniques, please refer to the work by Cai and Knight.

	M	P	R	F
x=w, y=b, z=g:	4	4/6	4/5	0.73
x=w, y=g, z=b:	1	1/6	1/5	0.18
x=b, y=w, z=g:	0	0/6	0/5	0
x=b, y=g, z=w:	0	0/6	0/5	0
x=g, y=w, z=b:	0	0/6	0/5	0
x=g, y=b, z=w:	2	2/6	2/5	0.36
Smatch score:				0.73

Table 2: Six ways of mapping variables of the two AMRs and their resulting F scores.

Smatch has become a standard evaluation metric for validating AMR parser training. It can also be directly applied to assess the semantic similarity between two AMR-parsed sentences, which is precisely the approach taken in this study. The similarity between the AMR representations of machine-generated English translations (first AMR) and those of the reference English translations (second AMR) of the same source sentences is measured using Smatch. The resulting Smatch scores are then used as the scores of the new machine translation evaluation metric (AMR), referred to here as the **AMR (Smatch) scores**.

One important task is to select an AMR parser that is not overly computationally expensive while maintaining high performance. Among the available models, the *AMR3-structbart-L* model (Drozdov et al., 2022) from the IBM transition-based AMR parser is chosen for its optimal performance-to-computational-load ratio. This transition-based AMR parser employs a neural aligner to learn word-node alignments—an essential component for transition-based AMR parsers—instead of relying on a traditional rule-based pipeline, which results in a substantial improvement in performance. It achieves an evaluation Smatch score of 0.826 (with beam size 10) on AMR3 test set, which is slightly lower than the top score of 0.844 obtained by the ensemble model, but it offers significantly faster

computation.
All model parameters remain at their default values except for a batch size of 64, with beam = 1 and n-best = 1, leading to greedy decoding. Although this may not produce the most accurate parsing, it is preferred due to the significantly lower computational cost of greedy decoding compared to more complex approaches, and it serves to establish a baseline performance for using AMR as an evaluation metric for machine translation.

2.3 Limitations

Using AMR as an evaluation metric for machine translation comes with limitations, which can be grouped into two categories: (1) limitations inherent to the AMR framework itself, and (2) limitations related to its application.

AMR is not a perfect semantic framework. To begin with, it lacks comprehensive coverage of all possible semantic relation frames (predicates), primarily due to limitations in PropBank (which covers only verb predicates). Many verb frames are missing, and numerous noun concepts cannot be naturally converted into verb predicates. For example, *professor* does not correspond to a verb frame like *profess-01* (Banarescu et al., 2013). More importantly, AMR's complete removal or omission of grammatical elements in sentences can result in the loss of certain aspects of meaning. AMR does not encode inflectional morphology for tense, and thus cannot distinguish between present, past, or future events. It also ignores distinctions introduced by articles, quantifiers, or number in noun phrases. However, such grammatical features can alter the truth conditions of propositions and are therefore crucial for precise semantic interpretation. Other semantic distinctions introduced by grammatical or functional elements —such as the difference between real and hypothetical events—are also ignored by AMR. For example, expressions like *could have been eaten*, *have been eaten*, and *if it is eaten* are all represented uniformly as ***eat-01***. In addition, AMR can strip away pragmatic cues from sentences. For instance, the use of active versus passive constructions may shift the topical focus, but AMR represents them identically, ignoring such nuances.

These inherent limitations of AMR may place it at a disadvantage position compared to neural network-based MT evaluation metrics, which are known for their ability to capture nuanced meanings.

The application of AMR as a machine translation (MT) evaluation metric also presents several challenges, particularly related to parsing accuracy and computational cost. Although the AMR parser used in this study is pretrained and publicly available (provided by IBM), generating accurate AMR graphs remains computationally intensive because the cost is not limited to training. Using the parser for inference can be expensive, depending on the decoding algorithm chosen. For example, using greedy decoding (beam size = 1) to parse a text of 1,976 sentences on CPU takes approximately half an hour. In contrast, applying beam search (with beam size > 1) to parse all 30 texts would require several days on CPU—resources that are not feasible for this study.
However, opting for simpler decoding comes at the expense of parsing accuracy. As will be discussed in the next section, the average Smatch score for all parsed sentences ranges between 0.6 and 0.7 under greedy decoding (Although a decrease on new data is typical.), in contrast to the parser's evaluation Smatch score of 0.826 (achieved with beam size 10 on AMR3 test set). These low scores reduce the reliability of AMR-based evaluation, particularly when parsing quality is inconsistent or when the method is applied to languages with limited AMR resources. For this reason, a single SOTA parser is used to parse all texts from the two language pairs considered in this study, and only language pairs with English as the target language are included, given that AMR is an English-centric framework and English AMR parsers perform the best (SoTA for English is 0.844 on AMR3, Chinese 0.63, German 0.737, Italian 0.761 and Spanish 0.771 (Lee et al., 2022)).
To balance parsing accuracy with computational cost, no optimal decoding is applied during parsing. Therefore, the goal of this paper is to establish a baseline for using AMR as a machine translation (MT) evaluation metric. This means that, with greater computational resources, AMR-based evaluation could poten-

tially yield better results than those presented here.

With the overall introduction and discussion of AMR as an MT evaluation metric complete, the next section focuses on its implementation, detailing the test data, MQM human ratings for the translations, the other MT evaluation metrics used for comparison, the comparison (correlation) methods and the performance of AMR relative to these established metrics.

3 Implementation and Results

To assess whether AMR is an effective MT evaluation metric, it must naturally be compared against a set of established, well-ranked MT evaluation metrics. The Conference on Machine Translation (WMT[4]), as the premier venue for machine translation research, hosts annual shared tasks and undoubtedly provides the most suitable data and resources for this purpose. It provides source texts and corresponding translations from over 10 different systems, along with human reference translations. In addition, more than 30 benchmark MT evaluation metrics are assessed and ranked using this data by comparing their evaluation scores against MQM human gold-standard evaluations. Another major advantage is that WMT also offers an analysis tool called MTME[5], which allows the integration of new metrics and enables consistent and robust comparison of the new metric with the established ones using their datasets and evaluation framework.

WMT hosts an annual Metrics Shared Task, with the most recent being the *WMT 2024 Metrics Shared Task*. Unfortunately, the 2024 results do not include any language pairs with English as the target language. Therefore, this study uses resources from the *WMT 2023 Metrics Shared Task*[6] for implementation. The metrics reported in 2023 largely overlap with those in 2024, with the exception of the newly introduced MetaMetrics-MT, which shares the top rank with XCOMET-Ensemble in 2024—XCOMET-Ensemble having been the highest-performing metric in 2023.

This section briefly introduces the data and other MT evaluation metrics used in the 2023 task, with a greater focus on their comparison methods employed to rank the participating evaluation metrics, followed by a presentation of the results and a discussion.

3.1 The WMT Test Data and Metric Set

2023 WMT Metrics Shared Task has two language pairs out of three with English as the target language: Hebrew-English (he→en) and Chinese - English (zh→en). The he→en dataset contains 1,910 source segments, while the zh→en dataset includes 1,976 segments. A segment refers to a single input unit for a translation system, which may consist of a paragraph, a full sentence or a partial sentence, such as a clause.

For he→en, translations are provided by 12 systems, and for zh→en by 15 systems, excluding human reference translations. These systems are listed in Table 4 with their human evaluations; for detailed descriptions of each system, refer to the WMT 2023 General MT Task report by Kocmi et al. (2023).

The test set with the source texts and their system outputs originate from the same year's (2023) WMT general machine translation task (except two translations from the system *NLLB200* model which is added in the 2023 Metrics Task for the effort to expand the pool of translations). WMT general task strives to provide dataset with good quality. The source texts were carefully selected to be recent, cross-domain, source-original, and human-refined, in order to reduce the likelihood of data contamination (i.e., overlap with MT training data) and to ensure high suitability for machine translation task (Kocmi et al., 2023).

A total of 32 benchmark metrics are reported in the WMT 2023 Metrics Shared Task. These include traditional metrics based on lexical overlap, such as BLEU and CHARF, as well as neural metrics that rely on embedding similarity like BERTSCORE, fine-tuned pre-trained models like COMET and XCOMET, large language models (LLMs) prompting like GEMBA-MQM or fully trained MT models like PRISM. Another important distinc-

[4]It was originally called the "Workshop on Machine Translation" before 2016. For details, see https://machinetranslate.org/wmt

[5]Machine Translation Metric Evaluation: https://github.com/google-research/mt-metrics-eval

[6]WMT 2023 Metrics Shared Task: https://wmt-metrics-task.github.io/

Language Pair	News	Social	Speech	User Reviews	Manuals
he→en	619/1558 (68/70)	n/a	n/a	201/352 (26/26)	n/a
zh→en	377/763 (38/38)	n/a	n/a	677/726 (127/127)	123/487 (14/14)

Table 3: Numbers of MQM-annotated segments per domain (number of docs in brackets).

	metric	broad category	supervised	ref. free
baselines	BLEU	lexical overlap		
	f200spBLEU	lexical overlap		
	chrF	lexical overlap		
	BERTscore	embedding similarity		
	BLEURT	fine-tuned metric	✓	
	COMET	fine-tuned metric	✓	
	COMETKIWI	fine-tuned metric	✓	✓
	docWMT22CometDA	fine-tuned metric	✓	
	docWMT22CometKiwiDA	fine-tuned metric	✓	✓
	MS-COMET-QE-22	fine-tuned metric	✓	✓
	prismRef	MT-model metric	✓	
	prismSrc	MT-model metric	✓	✓
	RANDOM-SYSNAME	random baseline		✓
	YiSi-1	embedding similarity		
primary submissions	CALIBRI-COMET22	fine-tuned metric	✓	
	CALIBRI-COMET22-QE	fine-tuned metric	✓	✓
	cometoid22-wmt22	fine-tuned metric	✓	✓
	eBLEU	embedding similarity		
	embed_llama	embedding similarity		
	GEMBA-MQM	LLM prompt-based metric		✓
	KG-BERTScore	embedding similarity		✓
	MATESE	fine-tuned metric	✓	
	mbr-metricx-qe	fine-tuned metric	✓	✓
	MEE4	lexical & embedding similarity		
	METRICX-23	fine-tuned metric	✓	
	METRICX-23-QE	fine-tuned metric	✓	✓
	mre-score-labse-regular	fine-tuned metric	✓	
	SESCORE	fine-tuned metric	✓	
	TOKENGRAM_F	lexical overlap		
	XCOMET-ENSEMBLE	fine-tuned metric	✓	
	XCOMET-QE-ENSEMBLE	fine-tuned metric	✓	✓
	XLSIM	fine-tuned metric	✓	

Figure 3: The 32 benchmark MT evaluation metrics of WMT 2023 Metrics Shared Task (Freitag et al., 2023)

tion among machine translation evaluation metrics is whether they are reference-based or reference-free. Traditional metrics typically assess translation quality by comparing machine-generated outputs against human reference translations. However, reference-free metrics have gained attention in recent years, partly due to concerns about the negative impact of low-quality or inconsistent reference translations (more about this in later sections). Notably, metrics such as GEMBA-MQM and XCOMET-QE-Ensemble are reference-free yet demonstrated strong performance in the WMT 2023 and 2024 Metrics Shared Task—ranking second or third overall in the evaluation.

The full list of all metrics is presented in Figure 3. For detailed descriptions of each metric, please refer to their respective papers, which are compiled in (Freitag et al., 2023).

3.2 MQM Human Metric

The WMT 2023 Metrics Shared Task also provides human ratings for a subset of system outputs, along with the human reference translations, proportionally sampled across docu-

English→German ↓

System	all	news	social	speech	user-reviews
refA	2.96	3.12	2.02	4.74	3.77
GPT4-5shot	3.72	4.00	2.41	6.51	4.60
ONLINE-W	3.95	2.69	2.62	5.90	7.13
ONLINE-B	4.71	4.35	3.14	5.96	7.85
ONLINE-Y	5.64	4.45	3.67	7.48	10.26
ONLINE-A	5.67	4.40	3.84	7.78	9.87
ONLINE-G	6.57	6.43	4.12	7.93	11.38
ONLINE-M	6.94	4.87	4.41	8.30	14.08
Lan-BridgeMT	8.67	7.99	5.55	9.72	15.78
ZengHuiMT	9.25	8.43	5.74	14.23	14.92
NLLB_Greedy	9.54	8.29	5.20	14.82	17.35
NLLB_MBR_BLEU	10.79	9.93	5.53	17.75	19.18
AIRC	14.23	14.32	8.34	20.34	23.45

Hebrew→English ↓

System	all	news	user-reviews
refA	1.17	1.28	0.86
GPT4-5shot	1.33	1.29	1.48
ONLINE-A	1.38	1.34	1.50
ONLINE-B	1.55	1.60	1.39
GTCOM_DLUT	1.89	1.85	1.99
UvA-LTL	1.92	1.80	2.30
ONLINE-G	2.06	2.06	2.04
ONLINE-Y	2.35	2.42	2.12
ZengHuiMT	2.38	2.33	2.53
Samsung_Research_Philippines	3.23	3.62	2.05
NLLB_MBR_BLEU	3.68	3.83	3.20
NLLB_Greedy	3.79	3.98	3.19
Lan-BridgeMT	3.79	3.81	3.74

Chinese→English ↓

System	all	news	manuals	user-reviews
Lan-BridgeMT	2.10	2.31	1.28	2.13
GPT4-5shot	2.31	2.26	2.01	2.39
Yishu	3.23	3.34	1.67	3.46
ONLINE-B	3.39	3.27	1.78	3.74
HW-TSC	3.40	3.40	1.83	3.68
ONLINE-A	3.79	2.90	1.83	4.63
ONLINE-Y	3.79	3.47	2.84	4.14
ONLINE-G	3.86	3.58	2.02	4.34
ONLINE-W	4.06	3.84	2.16	4.53
ZengHuiMT	4.23	4.05	2.84	4.59
IOL_Research	4.59	3.60	1.85	5.63
refA	4.83	5.04	5.17	4.65
ONLINE-M	5.43	4.71	2.98	6.28
ANVITA	6.08	5.17	2.97	7.15
NLLB_MBR_BLEU	6.36	6.57	3.39	6.78
NLLB_Greedy	6.57	6.70	2.95	7.16

Table 4: MQM human evaluations for generalMT2023. Lower MQM average error counts represent higher MT quality. Systems above any solid line are significantly better than those below, based on all domains with $p < 0.05$ (Freitag et al., 2023).

ments and domains. Ratings are assigned at the segment level, and the overall system score is calculated as the average rating across all segments for that system. These evaluations were conducted by professional translators or experienced native-speaking raters (8 for zh→en and 4 for he→en), who had access to full document context and followed a rigorous error-based evaluation framework known as Multidimensional Quality Metrics (MQM). The MQM framework explicitly categorizes translation errors by type (e.g., accuracy, fluency, style, non-translation and etc.) and severity (major, minor and neutral), organizing them into a hierarchical structure with different weights (Freitag et al., 2021). In this way, it avoids the issues associated with scalar ratings, where raters assign scores based on subjective judgments and implicit weighting schemes that may not reflect the priorities of the specific application. As shown in the study by Google research (Freitag et al., 2021), human evaluations collected using the MQM scoring scheme demonstrate a clear preference for human reference translations over machine-generated ones, and they correlate better with most MT evaluation metrics than the crowd-sourced scalar human scores used in WMT 2020. For this reason, starting in 2021, WMT adopted the MQM scoring scheme for human evaluation.

In WMT 2023, human MQM scores continue to show a preference for human reference translations over system outputs. For example, in both the English-German (en→de) and Hebrew-English (he→en) language pairs, the reference translations receive the lowest MQM scores which indicate the fewest errors (see Table 4). However, a discrepancy is observed in the Chinese-English (zh→en) results, where the reference translation ranks only sixth out of eight when ranked by statistically significant differences in MQM scores, as shown in Table 4. This is an indication of the poor quality of the reference translation of Chinese to English. In the WMT 2023 Metrics Shared Task results paper (Freitag et al., 2023), an entire section (Section 8) is devoted to discussing the substantial impact that low-quality reference translations can have on the assessment of MT evaluation metrics. The authors illustrate this

by comparing the correlation between poor reference translations and human MQM scores with the correlation between high-quality synthetic reference translations and human MQM scores. For the Chinese - English language pair, the correlation with human ratings increases dramatically when synthetic references are used, whereas for English-German, the difference is negligible. The synthetic references are created by selecting, for each segment, the candidate translation (from among all system outputs and the original reference) with the fewest errors—that is, the lowest MQM score. In this paper, for Chinese-English language pair, both the original and synthetic reference translations are used to produce results, in order to illustrate the importance of reference quality in reference-based MT evaluation. However, despite the potential bias in synthetic references, their ability to mitigate the impact of low-quality original references leads to their use as the main results in this paper[7].

3.3 AMR Integration and Metrics Comparison

In order to establish a ranking that reflects a metric's performance across a range of settings and applications, the WMT 2023 Metrics Task uses three correlation measurements—*Pearson*, *accuracy*, and accuracy with tie calibration (*acc-t* for short)—and combines the results into ten tasks, defined by the dimensions of language pairs, granularity, and correlation statistic. Since this paper examines only two of the three language pairs, the total number of tasks is reduced to seven, as shown in Table 5. It also computes a global weighted average correlation score for each metric across all tasks marked as "avg-corr".

Accuracy was proposed by (Kocmi et al., 2021) to measure how reliably an MT evaluation metric ranks systems in pairwise comparisons (as task 1) using human ratings as the gold standard. For each system pair, the differences in metric scores (metric Δ) and in human ratings (human Δ) are computed. *Accuracy* is then defined as the proportion of pairs where the metric and human score differences have the same sign—that is, when both agree on which system performs better—relative to the total number of pair comparisons across all systems, as shown below:

$$\text{Accuracy} = \frac{|\,\text{sign}(\text{metric }\Delta) = \text{sign}(\text{human }\Delta)|}{|\text{All System Pairs}|} \quad (1)$$

And it is done over data across both language pairs.

accu-t, proposed by (Deutsch et al., 2023), is used for segment-level pairwise ranking (as task 4, 7). It extends the *accuracy* metric by accounting for ties in either human ratings or metric scores. If both systems are judged equal by humans or by the metric, the comparison contributes a half weight (0.5) instead of a full weight. This adjustment provides a fairer evaluation when ties are frequent. The *accu-t* score for a system pair is computed by averaging the weighted results across all segment pairs.

In addition, ***Pearson*** correlation is used at both the system and segment levels (as task 2, 3, 5, 6) to compare metrics. For system-level evaluation, each metric provides a single score per system, which is then correlated with the corresponding MQM ratings. For segment-level evaluation, the system × segment score matrix is flattened into a single vector before computing the correlation.

When computing the **global average correlation score** (*avg-corr*) for each metric across tasks, **task weights** are applied: all tasks have a weight of 1, except system-level *accuracy*, which is weighted 2 as it combines data from both language pairs. They are computed by first mapping *Pearson* correlations

Task	Language	Granularity	Correlation	Weight
1	all	system	accuracy	2
2	he→en	system	Pearson	1
3	he→en	segment	Pearson	1
4	he→en	segment	acc-t	1
5	zh→en	system	Pearson	1
6	zh→en	segment	Pearson	1
7	zh→en	segment	acc-t	1

Table 5: The tasks and their weights for metrics ranking results

[7]Note that although about 10% of the synthetic reference come from the original reference, given that most are drawn from system outputs, the synthetic reference created this way is not ideal as it can introduce bias toward those systems.

from their natural range of [−1, 1] to the [0, 1] range of *accuracy* scores, and then taking a weighted average of the results. However, as noted in the WMT 2023 report, this weighting scheme has limitations. Since system-level correlation scores generally have higher means and standard deviations than segment-level scores, equal averaging across levels effectively upweights system-level contributions. In addition, the scheme assumes independence between tasks, which may not hold for system- and segment-level correlation scores derived from the same data. Nevertheless, WMT also experimented with different weightings and found minimal differences in results, so this paper follows their task weighting when computing the global average correlations.

In the WMT results, for **rank assignment**, an additional step of significance clustering was performed based on pairwise metric correlation score differences. This was done using 1,000 bootstrap resampling runs with a significance threshold of p=0.05, resulting in nine distinct clusters (i.e., nine ranks). Due to the high computational cost and the limited resources available, this paper does not replicate this step and instead ranks the metrics directly based on their raw correlation scores. However, for reference, the correlation ranges corresponding to each cluster are reported here: rank 1 : 0.825, rank 2 : 0.800-0.808, rank 3 : 0.772-0.788, rank 4 : 0.754-0.772, rank 5 : 0.742-0.744, rank 6 : 0.719, rank 7 : 0.692-0.704, rank 8 : 0.529, rank 9 : 0.455.

To accommodate the WMT framework, AMR Smatch scores are computed at both the system and segment levels and provided to WMT's new metric integration tool, MTME. At the segment level, Smatch scores are calculated by comparing the total and matching triplets for each segment pair between the system output and its reference. At the system level, the total triplets and matching triplets are aggregated across all segments before computing the score. For the Chinese ‑ English (zh→en) language pair, results are computed using both the low-quality original reference translations and the high-quality synthetic reference translations for comparison.

The distributions of AMR Smatch scores at both the system and segment levels are shown in Figures 4 and 5. For Hebrew to English, all systems perform similarly, with scores ranging from 0.65 to 0.71 and an average of 0.68. Notably, GPT4-5shot achieves the highest score (0.71), even outperforming the alternative human reference A. In contrast, the Chinese to English results display much greater variation, with scores ranging from 0.47 to 0.76. The average score is 0.58 when using human reference A, but rises to 0.63 with the synthetic reference. Unlike the Hebrew to English case, the reference translations achieve the highest score overall (despite the poor quality of reference A), followed closely by HW-TSC, which ranks highest among the system outputs. A key observation is that Smatch scores based on the synthetic reference in Chinese-English translations are consistently higher than those based on the low-quality human reference A across all systems, indicating that system outputs align more closely with the higher-quality reference favored by human judgments. At the segment level, the results mirror the system-level trends: Hebrew‑English translations generally align better with the references than Chinese-English translations, and the use of the synthetic reference for Chinese-English improves agreement overall.

With the AMR MT quality evaluation scores (Smatch scores) computed, MT evaluation metric rankings that include AMR as a new metric can be determined using WMT's correlation methods: *accuracy*, *accuracy-t*, and *Pearson*. The next section presents these ranking results, followed by a discussion.

3.4 Results and Discussion

The rankings of MT evaluation metrics with AMR included are shown in Table 6 and Table 7. Table 6 presents the results where AMR Smatch scores are computed using the low-quality human referenceA, while Table 7 shows the results based on high-quality synthetic references. As expected, replacing the low-quality human references with synthetic reference translations for zh→en—whose quality aligns more closely with human MQM ratings—significantly boosts the overall correlation between AMR scores and human ratings on the system outputs. Specifically, AMR's global ranking as an MT evaluation metric improves from 27th to 23rd, with the

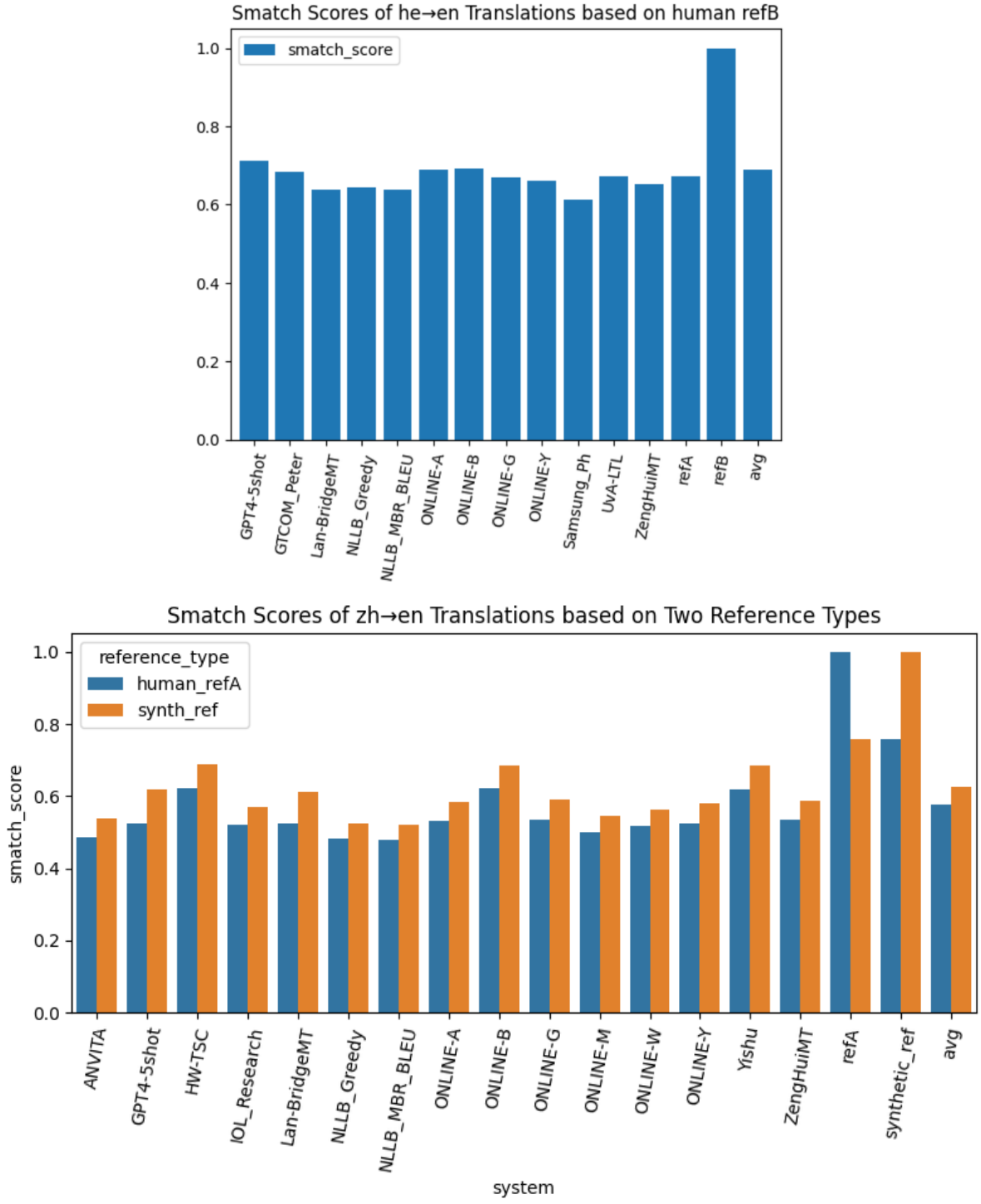


Figure 4: System level Smatch scores for both language pairs

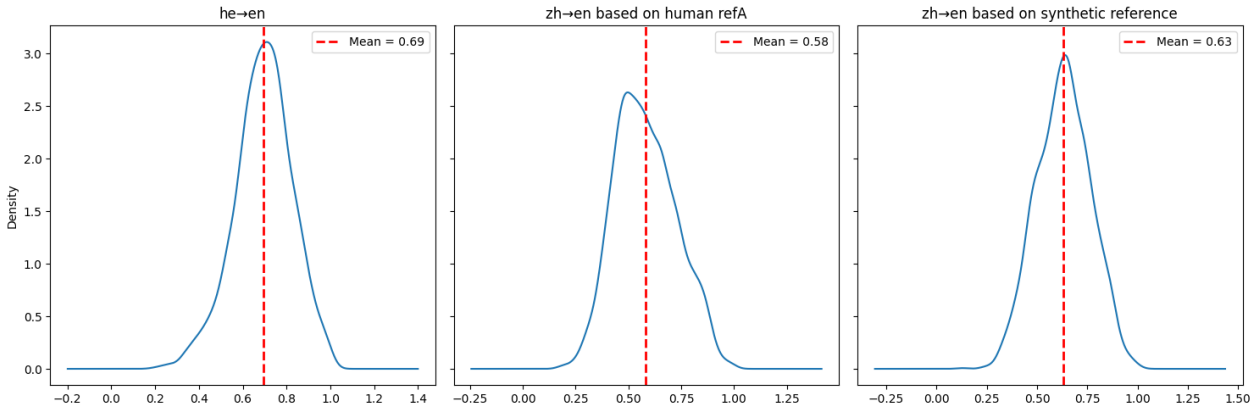


Figure 5: Segment level Smatch scores for both language pairs

global average correlation (*avg-corr*) increasing from 0.686 to 0.722. All three zh→en tasks (Tasks 5, 6, and 7) contribute to this gain, but the largest impact comes from Task 7, the segment-level *accu-t* ranking, which jumps from 29th place to 5th—outperforming most neural MT metrics.

This result demonstrates that, with high-quality human reference translations, using AMR as a metric for MT evaluation performs better than all string-based metrics and even surpasses some embedding-based metrics, such as *eBLEU* and *embed_llama*, as well as fine-tuned metrics like *XLsim*. Its performance is also comparable to strong neural baselines like *primRef* and *BERTScore* and *MS-COMET-QE-22*. Based on the nine-cluster significance ranking, AMR would fall between the 5th and 6th clusters, with a global average correlation score of 0.722. These results suggest that AMR effectively captures semantic information encoded in the meaning representation structures—information that purely lexical-based metrics fail to account for.

However, the gap between AMR and top-performing neural metrics is significant. As discussed in Section 2.3, the omission of grammatical features in the AMR framework causes the semantic representations to lose important nuances of meaning. Additionally, errors introduced by imperfect AMR parsers further limit its accuracy. These factors make it challenging for AMR-based evaluation to compete with top-notch fine-tuned neural metrics such as *XCOMET* and *MetricX*.

The advantages of AMR over fine-tuned neural metrics are its lower computational cost and its relative domain-independence. In other words, if a metric better than BLEU is desired, AMR can serve as a viable alternative to embedding-based MT evaluation metrics. Notably, the reported performance is achieved using the most basic parsing settings, despite many yet-to-resolve issues in the current AMR framework. With optimized parsing parameters and further enhancements to the framework, AMR's effectiveness as a machine translation evaluation metric could be significantly improved.

What's more, AMR, like many other metrics, has the potential to become reference-free, allowing system translations to be compared directly with the source text rather than reference translations, thereby avoiding reference bias. This can be achieved either through a multilingual AMR parser or by employing an alignment method between AMR concepts across different languages. In addition, with the availability of document-level AMR annotations, evaluations could incorporate cross-sentence semantic relations, enabling more comprehensive and context-aware assessments of translation quality.

Future work is expected to explore all these promising directions of research.

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language pair:			he→en,zh→en		he→en		he→en		he→en		zh→en		zh→en		zh→en	
level:			sys		sys		seg		seg		sys		seg		seg	
correlation function:			accuracy		pearson		pearson		acc-t		pearson		pearson		acc-t	
metric	avg-corr		task1		task2		task3		task4		task5		task6		task7	
XCOMET-Ensemble	**1**	**0.815**	3	0.923	3	0.950	**1**	**0.556**	**1**	**0.586**	8	0.927	**1**	**0.650**	**1**	**0.543**
MetricX-23	2	0.800	4	0.907	12	0.910	2	0.548	2	0.577	16	0.873	4	0.625	4	0.531
XCOMET-QE-Ensemble*	3	0.793	6	0.891	13	0.909	4	0.498	5	0.554	13	0.892	3	0.647	3	0.533
GEMBA-MQM*	4	0.790	**1**	**0.923**	5	0.939	7	0.401	4	0.564	**1**	**0.991**	9	0.449	7	0.522
MetricX-23-QE*	5	0.787	8	0.880	23	0.858	3	0.520	3	0.564	18	0.859	2	0.647	5	0.527
MaTESe	6	0.782	2	0.923	15	0.906	5	0.459	7	0.550	14	0.889	6	0.511	24	0.479
mbr-metricx-qe*	7	0.773	19	0.858	10	0.915	6	0.411	6	0.553	6	0.936	7	0.489	2	0.537
CometKiwi*	8	0.768	10	0.874	22	0.860	12	0.387	8	0.544	3	0.963	11	0.442	6	0.525
KG-BERTScore*	9	0.765	16	0.863	14	0.908	14	0.382	9	0.537	4	0.962	12	0.430	9	0.516
COMET	10	0.764	9	0.874	4	0.940	8	0.401	10	0.532	12	0.898	13	0.396	11	0.514
cometoid22-wmt22*	11	0.760	14	0.869	26	0.839	17	0.365	15	0.515	5	0.940	8	0.479	10	0.515
Calibri-COMET22	12	0.759	5	0.896	7	0.930	9	0.401	16	0.515	17	0.863	14	0.396	26	0.474
BLEURT-20	13	0.757	15	0.863	6	0.937	15	0.382	14	0.519	15	0.880	16	0.378	8	0.518
sescoreX	14	0.753	17	0.863	17	0.901	13	0.385	22	0.484	22	0.797	5	0.536	14	0.499
docWMT22CometKiwiDA*	15	0.750	12	0.874	16	0.906	22	0.286	21	0.489	2	0.965	15	0.387	17	0.493
docWMT22CometDA	16	0.750	11	0.874	8	0.922	18	0.339	20	0.497	9	0.907	18	0.353	18	0.493
mre-score-labse-regular	17	0.746	7	0.885	**1**	**0.958**	16	0.378	13	0.522	10	0.903	25	0.145	22	0.481
YiSi-1	18	0.745	18	0.858	9	0.917	10	0.395	11	0.529	19	0.823	19	0.290	12	0.504
Calibri-COMET22-QE*	19	0.743	23	0.836	28	0.778	11	0.395	18	0.506	7	0.934	10	0.443	19	0.491
BERTscore	20	0.736	13	0.869	18	0.895	19	0.335	17	0.515	20	0.810	20	0.236	15	0.499
MS-COMET-QE-22*	21	0.729	21	0.847	31	0.721	21	0.295	19	0.498	11	0.901	17	0.367	16	0.498
prismRef	22	0.727	25	0.836	2	0.956	20	0.319	12	0.528	27	0.762	23	0.183	13	0.504
XLsim	23	0.705	22	0.842	19	0.887	25	0.233	23	0.480	23	0.796	27	0.111	30	0.464
embed_llama	24	0.694	24	0.836	25	0.841	30	0.215	32	0.430	25	0.785	24	0.161	31	0.447
tokengram_F	25	0.692	28	0.803	21	0.878	27	0.226	24	0.461	24	0.795	31	0.060	21	0.485
MEE4	26	0.691	26	0.809	20	0.879	23	0.256	31	0.441	28	0.743	29	0.105	23	0.480
AMR	27	0.686	27	0.803	24	0.846	24	0.245	26	0.454	31	0.605	22	0.217	29	0.468
f200spBLEU	28	0.685	29	0.798	27	0.805	26	0.230	27	0.447	26	0.772	28	0.108	25	0.476
eBLEU	29	0.683	20	0.852	11	0.911	32	0.131	28	0.445	30	0.727	33	-0.084	27	0.473
chrF	30	0.682	31	0.787	29	0.776	28	0.221	25	0.460	21	0.809	30	0.063	20	0.485
BLEU	31	0.676	30	0.787	30	0.769	29	0.220	29	0.442	29	0.734	26	0.119	28	0.472
Random-sysname*	32	0.520	32	0.585	32	0.209	33	0.041	33	0.428	32	0.093	32	0.018	33	0.381
prismSrc*	33	0.462	33	0.432	33	-0.017	31	0.140	30	0.441	33	-0.406	21	0.223	32	0.421

Table 6: MT evaluation metrics ranking with AMR included. AMR Smatch scores are computed based on human reference A.

language pair:			he→en,zh→en		he→en		he→en		he→en		zh→en		zh→en		zh→en	
level:			sys		sys		seg		seg		sys		seg		seg	
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mbr-metricx-qe*	7	0.773	19	0.858	10	0.915	6	0.411	6	0.553	6	0.936	7	0.489	2	0.537
CometKiwi*	8	0.768	10	0.874	22	0.860	12	0.387	8	0.544	3	0.963	11	0.442	7	0.525
KG-BERTScore*	9	0.765	16	0.863	14	0.908	14	0.382	9	0.537	4	0.962	12	0.430	10	0.516
COMET	10	0.764	9	0.874	4	0.940	8	0.401	10	0.532	12	0.898	13	0.396	12	0.514
cometoid22-wmt22*	11	0.760	14	0.869	26	0.839	17	0.365	15	0.515	5	0.940	8	0.479	11	0.515
Calibri-COMET22	12	0.759	5	0.896	7	0.930	9	0.401	16	0.515	17	0.863	14	0.396	27	0.474
BLEURT-20	13	0.757	15	0.863	6	0.937	15	0.382	14	0.519	15	0.880	16	0.378	9	0.518
sescoreX	14	0.753	17	0.863	17	0.901	13	0.385	22	0.484	22	0.797	5	0.536	15	0.499
docWMT22CometKiwiDA*	15	0.750	12	0.874	16	0.906	22	0.286	21	0.489	2	0.965	15	0.387	18	0.493
docWMT22CometDA	16	0.750	11	0.874	8	0.922	18	0.339	20	0.497	9	0.907	18	0.353	19	0.493
mre-score-labse-regular	17	0.746	7	0.885	**1**	**0.958**	16	0.378	13	0.522	10	0.903	25	0.145	23	0.481
YiSi-1	18	0.745	18	0.858	9	0.917	10	0.395	11	0.529	19	0.823	20	0.290	13	0.504
Calibri-COMET22-QE*	19	0.743	24	0.836	28	0.778	11	0.395	18	0.506	7	0.934	10	0.443	20	0.491
BERTscore	20	0.736	13	0.869	18	0.895	19	0.335	17	0.515	20	0.810	21	0.236	16	0.499
MS-COMET-QE-22*	21	0.729	22	0.847	31	0.721	21	0.295	19	0.498	11	0.901	17	0.367	17	0.498
prismRef	22	0.727	26	0.836	2	0.956	20	0.319	12	0.528	27	0.762	23	0.183	14	0.504
AMR	23	0.722	20	0.852	24	0.846	24	0.245	26	0.454	28	0.753	19	0.338	5	0.528
XLsim	24	0.705	23	0.842	19	0.887	25	0.233	23	0.480	23	0.796	27	0.111	30	0.464
embed_llama	25	0.694	25	0.836	25	0.841	30	0.215	32	0.430	25	0.785	24	0.161	31	0.447
tokengram_F	26	0.692	28	0.803	21	0.878	27	0.226	24	0.461	24	0.795	31	0.060	22	0.485
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eBLEU	29	0.683	21	0.852	11	0.911	32	0.131	28	0.445	31	0.727	33	-0.084	28	0.473
chrF	30	0.682	31	0.787	29	0.776	28	0.221	25	0.460	21	0.809	30	0.063	21	0.485
BLEU	31	0.676	30	0.787	30	0.769	29	0.220	29	0.442	30	0.734	26	0.119	29	0.472
Random-sysname*	32	0.520	32	0.585	32	0.209	33	0.041	33	0.428	32	0.093	32	0.018	33	0.381
prismSrc*	33	0.462	33	0.432	33	-0.017	31	0.140	30	0.441	33	-0.406	22	0.223	32	0.421

Table 7: MT evaluation metrics ranking with AMR included. AMR Smatch scores are computed based on synthetic reference.

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